Calphad 66 (2019) 101646

Contents lists available at ScienceDirect

Calphad

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic re-assessment of the Fe-Tm and Fe-Ho binary systems

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ARTICLE INFO

Keywords:
Fe-Tm
Fe-Ho
Phase diagram
Thermodynamics
CALPHAD

ABSTRACT

Thermodynamic re-assessment of the Fe-Tm and Fe-Ho binary systems were carried out with the help of the CALPHAD method based on the previous optimizations and the critical review of the available experimental information in the published literature. The substitutional solution model was used to describe liquid phase and fcc-Fe, bcc-Fe, hcp-Tm, bcc-Ho and hcp-Ho solid solution phases and their excess terms of Gibbs energies were expressed with the Redlich-Kister polynomial. The intermetallic compounds, $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm, Fe_2Tm, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$, Fe_3Ho and Fe_2Ho, were treated as stoichiometric compounds considering the experimental heat capacity of two intermetallic compounds (Fe_2Tm and Fe_2Ho) in the low temperature range. Self-consistent thermodynamic parameters to describe the Gibbs energies of various phases in the Fe-Tm and Fe-Ho binary systems were obtained finally. The calculated results in this work are in good agreement with available phase equilibira data and thermodynamic data.

1. Introduction

Nd-Fe-B permanent magnets with outstanding magnetic properties have been used widely in many industrial applications (e.g. communication, transportation, aerospace and wind power) [1–4]. Nd-Fe-B permanent magnets with heavy rare earth (RE) metals Dy and Tb show higher magnetic energy product, higher coercivity and better temperature stability [5,6]. The imbalance of rare earth metals supply and demand results in the unreasonable utilization of rare earth resources. Nd-Fe-B permanent magnets with other heavy rare earth metals (e.g. Tm, Ho) were studied experimentally [7–9]. The addition Ho to Nd-Fe-B magnets promotes the growth of $Nd_2Fe_{14}B$ main phase to result in the uniform distribution of Nd-rich phase and the grain refinement of the magnets. The effective Ho content on coercivity enhancement of 0.64 kOe/wt.% was obtained in Nd-Fe-B magnets with 4 wt% Ho [7,8]. In addition, the addition of Tm would improve the saturation magnetization, thermal stability and corrosion resistant of Nd-Fe-B magnets compared with the magnets without heavy-rare-earth elements (e.g. Dy, Tb) [9]. In order to understand better the effect of rare earth metals Tm and Ho on phase formation, microstructure and magnetic properties of Nd-Fe-B permanent magnets, phase equilibria and thermodynamic information of the related RE-Fe-B alloy systems are indispensable.

Recently, the Fe-RE binary systems as the sub-binary systems in the RE-Fe-B ternary systems including Fe-Nd, Fe-Pr, Fe-Sm, Fe-Gd, Fe-Dy and Fe-Tb binary system were assessed thermodynamically in our previous work [10–13]. The Fe-Tm binary system was optimized by Kardellass et al. [14] and Konar et al. [16], while the Fe-Ho binary system was assessed by Kardellass et al. [15] and Konar et al. [16]. The calculated Fe-Tm binary phase diagram by Kardellass et al. [14] using the CALPHAD method was in good agreement with the experimental results determined by the Kolesnichenko et al. [17] in the high temperature range (above 1200 K). However, the calculated temperature of the eutectic reaction, $L \leftrightarrow Fe_2Tm + \alpha\text{-Tm}$, is lower (28 K) than the experimental data [17] and some intermetallic compounds (e.g. $Fe_{23}Tm_6$, Fe_3Tm and Fe_2Tm) was decomposed abnormally at low temperature. On the other hand, the calculated Fe-Ho binary phase diagram by Kardellass et al. [15] is consistent with the experimental results measured by Roe et al. [18]. However, the calculated results [15] show that two intermetallic compounds $Fe_{17}Ho_2$ and $Fe_{23}Ho_6$ were decomposed at low temperature, which is obviously inconsistent with the experimental results [18]. Recently, Konar et al. [16] re-optimized the Fe-Tm and Fe-Ho binary systems based on the new experimental data determined by Ivanov et al. [19] using modified quasichemical model to describe liquid phase. Although the calculated binary phase diagrams and thermodynamically properties are in good agreement with the experimental data [17–19], the Fe-Tm and Fe-Ho binary systems are still necessary to be re-assessed thermodynamically, considering the compatibility of thermodynamic database of multicomponent RE-Fe-B alloy systems.

Therefore, the purpose of the present work is to evaluate the

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https://doi.org/10.1016/j.calphad.2019.101646
Received 5 April 2019; Received in revised form 17 July 2019; Accepted 19 July 2019
Available online 31 July 2019
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Table 1
Crystallographic data of stable phases in the Fe-Tm and Fe-Ho binary systems.

Phase	Prototype	Space Group	Lattice parameters		References
			a (nm)	c (nm)	
δ-Fe	W	$Im\bar{3}m$	0.2932	–	[20]
γ-Fe	Cu	$Fm\bar{3}m$	0.3647	–	[20]
α-Fe	W	$Im\bar{3}m$	0.2867	–	[20]
α-Tm	Mg	$P6_3/mcm$	0.3538	0.5554	[20]
α-Ho	Mg	$P6_3/mmc$	0.3578	0.5614	[20]
β-Ho	W	$R\bar{3}m$	0.3962	–	[20]
$Fe_{17}Tm_2$	Th_2Ni_{17}	$P6_3/mcm$	0.8301	0.8298	[22]
$Fe_{23}Tm_6$	Th_6Mn_{23}	Fm_3m	1.1980	–	[22]
Fe_3Tm	$NiPu_3$	$R\bar{3}m$	–	–	[22]
Fe_2Tm	$MgCu_2$	$Fd\bar{3}m$	0.7240	–	[22]
$Fe_{17}Ho_2$	Th_2Ni_{17}	$P6_3/mcm$	0.8440	0.8320	[31]
$Fe_{23}Ho_6$	Th_6Mn_{23}	Fm_3m	1.2040	–	[31]
Fe_3Ho	$NiPu_3$	$R\bar{3}m$	0.5084	2.5450	[31]
Fe_2Ho	$MgCu_2$	$Fd\bar{3}m$	0.7280	–	[31]

experimental phase diagram and thermodynamic data of the Fe-Tm and Fe-Ho binary systems and then to obtain consistent and reliable thermodynamic parameters of the stable phases in these two binary systems.

2. Experimental information

2.1. The Fe-Tm binary system

The Fe-Tm binary phase diagram was determined by Kolesnichenko et al. [17] by means of X-ray powder diffraction (XRD), differential thermal analysis (DTA) and metallographic technique. Massalski [20] and Okamoto [21] reviewed the Fe-Tm binary system. Four intermetallic compounds, $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm and Fe_2Tm, were found. $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, and Fe_3Tm are formed through three peritectic reactions at 1573 K, 1543 K, and 1551 K, respectively, while Fe_2Tm is formed by melting congruently at 1573 K. The temperatures and compositions of two eutectic reactions, L ↔ Fe_2Tm + α-Tm and L ↔ $Fe_{23}Tm_6$ + $Fe_{17}Tm_2$, were determined to be 1210 K and 73.0 at. % Tm, 1528 K and 17.0 at. % Tm, respectively. The composition ranges of $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm and Fe_2Tm were not studied. Based on the review of Fe-lanthanide binary systems [22], the intermetallic compounds in the Fe-Tm binary system are considered as stoichiometric compounds in this work. There are no reports of the solid solubility of Fe in hcp-Tm (α-Tm) and Tm in fcc-Fe (γ-Fe) and bcc-Fe (δ-Fe and α-Fe) in the published literature. Table 1 summarizes the crystal structure information of stable phases in the Fe-Tm binary system.

The heat capacity of Fe_2Tm was determined experimentally by Germano et al. [23] using adiabatic calorimetry in the low temperature range (8–300 K). Gozzi et al. [24] measured the enthalpy of formation of $Fe_{17}Tm_2$ at 959 K to be −4.17 ± 0.04 kJ/mol-atom by the electromotice force method (EMF). Meschel et al. [25] determined the enthalpy of formation of Fe_2Tm at 1373 K to be −2.2 ± 2.8 kJ/mol-atom by the high temperature direct synthesis calorimetry. In addition, using density functional theory, the enthalpies of formation of $FeTm_2$, Fe_3Tm, Fe_6Tm_{23} and $Fe_{17}Tm_2$ compounds were calculated by the Materials Project [26] to be 13.465 kJ/mol-atom, −1.1053 kJ/mol-atom, −0.797 kJ/mol-atom and −0.2496 kJ/mol-atom, respectively, while were also calculated by the Open Quantum Database (OQMD) [27] to be −13.463 kJ/mol-atom, −1.1057 kJ/mol-atom, −0.798 kJ/mol-atom and −0.2498 kJ/mol-atom.

2.2. The Fe-Ho binary system

The Fe-Ho binary phase diagram was investigated experimentally by Roe et al. [18] using XRD, DTA and metallographic technique. Okamoto [21] reviewed this binary system. Four intermetallic compounds, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$, Fe_3Ho and Fe_2Ho, were confirmed. There are four eutectic reactions (L ↔ $Fe_{17}Ho_2$ + γ-Fe at 1611 K and 8.3 at.% Ho, L ↔ $Fe_{17}Ho_2$ + $Fe_{23}Ho_6$ at 1585 K and 17.8 at.% Ho, L ↔ Fe_2Ho + Fe_3Ho at 1561 K and 34.9 at.% Ho, and L ↔ α-Ho + Fe_2Ho at 1148 K and 63.4 at.% Ho). $Fe_{17}Ho_2$, $Fe_{23}Ho_6$ and Fe_2Ho melt congruently at 1616 K, 1604 K and 1563 K, respectively, while Fe_3Ho is formed by the peritectic reaction (L + $Fe_{23}Ho_6$ ↔ Fe_3Ho) at 1566 K. No experimental solubility of $Fe_{17}Ho_2$, $Fe_{23}Ho_6$, Fe_3Ho and Fe_2Ho is available. There are no reports of the solubility of Fe in hcp-Ho (α-Ho) and bcc-Ho (β-Ho) or Ho in fcc-Fe (γ-Fe) and bcc-Fe (α-Fe and δ-Fe) in the reported literature. The experimental data including liquidus and invariant reactions determined by Roe et al. [18] were accepted in this work. The crystal structure information of stable phases in the Fe-Ho binary system was given in Table 1.

The heat capacity of Fe_2Ho was determined by Germano et al. [23] in the low temperature range (8–300 K) by means of the isoperibolic calorimeter. Gozzi et al. [24] determined the enthalpies of formation of $Fe_{17}Ho_2$ and $Fe_{23}Ho_6$ to be −4.32 ± 0.02 kJ/mol-atom at 1066 K and −6.7 kJ/mol-atom at 298.15 K by EMF, respectively. Meschel et al.

Table 2
Thermodynamic parameters of the Fe-Tm binary system.

Phase	Thermodynamic parameters	Reference
Liquid	${}^{(0)}L^{Liq} = -37000 + 15.844T$ ${}^{(1)}L^{Liq} = -16427 + 11.011T$ ${}^{(2)}L^{Liq} = -8539.50 + 9.955T$	This work
fcc-A1 (γ-Fe)	${}^{0}G_{Fe}^{fcc}$ cited from SGTE database	[28]
bcc-A2 (α-Fe, δ-Fe)	${}^{0}G_{Fe}^{bcc}$ cited from SGTE database	[28]
hcp-A3 (α-Tm)	${}^{0}L_{Fe,Tm}^{hcp} = 50000$	This work
$Fe_{17}Tm_2$	$G_m^{Fe_{17}Tm_2} = -11500 + 5.255T + 0.8947\,{}^{0}G_{Fe}^{bcc} + 0.1053\,{}^{0}G_{Tm}^{hcp}$	This work
	$T_C^{Fe_{17}Tm_2} = 280$ $\beta_0^{Fe_{17}Tm_2} = 1.07$	[32]
$Fe_{23}Tm_6$	$G_m^{Fe_{23}Tm_6} = -13850 + 4.95T + 0.7931\,{}^{0}G_{Fe}^{bcc} + 0.2069\,{}^{0}G_{Tm}^{hcp}$	This work
	$T_C^{Fe_{23}Tm_6} = 475 \cdot \beta_0^{Fe_{23}Tm_6} = 0.62$	[32]
Fe_3Tm	$G_m^{Fe_3Tm} = -14800 + 4.850T + 0.75\,{}^{0}G_{Fe}^{bcc} + 0.25\,{}^{0}G_{Tm}^{hcp}$	This work
	$T_C^{Fe_3Tm} = 537 \cdot \beta_0^{Fe_3Tm} = 0.40$	[32]
Fe_2Tm	$G_m^{Fe_2Tm} = \begin{cases} -15434.86 + 0.2T - 7\times10^{-4}T^3 + 1.9\times10^{-6}T^4 & (T<120) \\ \left.\begin{matrix} -15500 + 21.458T - 3T\ln(T) + 0.004T^2 \\ -25000T^{-1} - 6\times10^{-7}T^3 + 0.6667\,{}^{0}G_{Fe}^{bcc} + 0.3333\,{}^{0}G_{Tm}^{hcp} \end{matrix}\right\} & (T<120) \end{cases}$	This work
	$T_C^{Fe_2Tm} = 599$ $\beta_0^{Fe_2Tm} = 0.87$	[32]

Table 3
Thermodynamic parameters of the Fe-Ho binary system.

Phase	Thermodynamic parameters	Reference
Liquid	${}^{(0)}L^{Liq} = -35377.26 + 8.740T$ ${}^{(1)}L^{Liq} = -14905.94 + 5.448T$	This work
fcc-A1 (γ-Fe)	${}^{0}G_{Fe}^{fcc}$ cited from SGTE database	[28]
bcc-A2 (α-Fe, β-Ho, δ-Fe)	${}^{(0)}L_{Fe,Ho}^{bcc} = 85000$	This work
hcp-A3 (α-Ho)	${}^{(0)}L_{Fe,Ho}^{hcp} = 31000$	This work
$Fe_{17}Ho_2$	$G_m^{Fe_{17}Ho_2} = -10200 + 2.818T + 0.8947\,{}^{0}G_{Fe}^{bcc} + 0.1053\,{}^{0}G_{Ho}^{hcp}$	This work
	$T_C^{Fe_{17}Ho_2} = 325 \cdot \beta_C^{Fe_{17}Ho_2} = 0.78$	[32]
$Fe_{23}Ho_6$	$G_m^{Fe_{23}Ho_6} = -12100 + 1.620T + 0.7931\,{}^{0}G_{Fe}^{bcc} + 0.2069\,{}^{0}G_{Ho}^{hcp}$	This work
	$T_C^{Fe_3Ho_6} = 530\ \beta_0^{Fe_3Ho_6} = 0.50$	[32]
Fe_3Ho	$G_m^{Fe_3Ho} = -11800 + 0.878T + 0.75\,{}^{0}G_{Fe}^{bcc} + 0.25\,{}^{0}G_{Ho}^{hcp}$	This work
	$T_C^{Fe_3Ho} = 571\ \beta_0^{Fe_3Ho} = 1.13$	[32]
Fe_2Ho	$G_m^{Fe_2Ho} = \begin{cases} -11974 - 8T - 7\times10^{-4}T^3 + 1.9\times10^{-6}T^4 & (T<120) \\ \left.\begin{matrix} -13500 + 28.880T - 4T\ln(T) + 0.001T^2 \\ +0.6667\,{}^{0}G_{Fe}^{bcc} + 0.3333\,{}^{0}G_{Ho}^{hcp} \end{matrix}\right\} & (T>120) \end{cases}$	This work
	$T_C^{Fe_2Ho} = 608\ \beta_0^{Fe_2Ho} = 1.84$	[32]

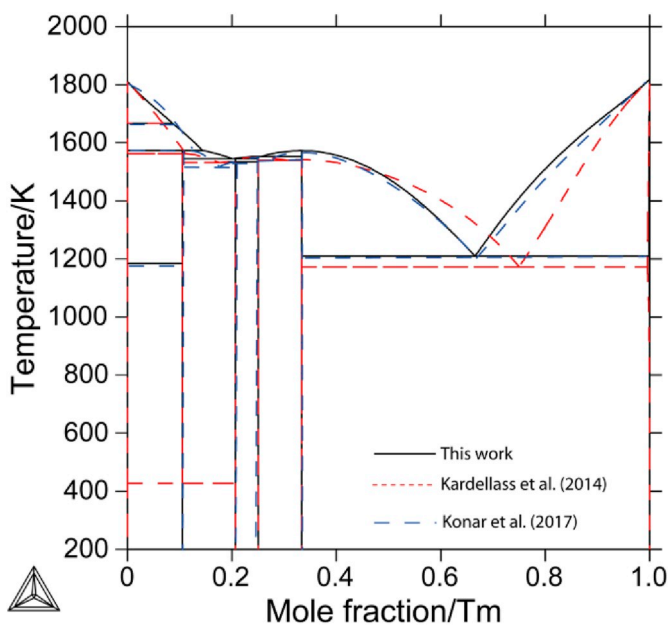

Fig. 1. Calculated phase diagram of the Fe-Tm binary system in this work with the calculated results [14,16].

[25] determined the enthalpy of formation of Fe_2Ho at 1373 K to be −2.6 ± 3.3 kJ/mol-atom by the high temperature direct synthesis calorimetry. In addition, the enthalpies of formation of Fe_2Ho, Fe_3Ho and $Fe_{17}Ho_2$ compounds were calculated by the Materials Project [26] to be −9.519 kJ/mol-atom, −8.289 kJ/mol-atom and −1.731 kJ/mol-atom, respectively, while were also calculated to be −9.514 kJ/mol-atom, −8.287 kJ/mol-atom and −1.732 kJ/mol-atom by the Open Quantum Database (OQMD) [27].

3. Thermodynamic modeling

3.1. Solid solution phases

Based on the random mixing of the constituent atoms, the solution phase φ (φ = liquid, bcc-Fe, fcc-Fe, hcp-Tm, bcc-Ho, hcp-Ho) in the Fe-Tm and Fe-Ho binary systems is described with the substitutional solution model. The molar Gibbs energy of the phase φ can be expressed as:

$$G_m^{\varphi} = \sum_{i=RE,Fe} {}^{0}G_i^{\varphi} x_i + RT \sum_{i=RE,Fe} x_i \ln x_i + {}^{ex}G_m^{\varphi} + {}^{mag}G_m^{\varphi} \tag{1}$$

$${}^{0}G_i^{\varphi}(T) = a + bT + cT\ln T + dT^2 + eT^3 fT^{-1} + gT^7 + hT^{-9} \tag{2}$$

$${}^{ex}G_m^{\varphi} = x_{Fe}x_{RE}\sum_{j=0}^{n}(A_j + B_jT)(x_{Fe} - x_{RE})^j \tag{3}$$

$${}^{mag}G_m^{\varphi} = RT\ln(\beta_0 + 1)g(\tau) \tag{4}$$

in which ${}^{0}G_i^{\varphi}(T)$ is the Gibbs energy of element i in the structure of the phase φ referred to the enthalpy of its stable state at 298.15 K and 1 bar. In this work, ${}^{0}G_i^{\varphi}(T)$ is taken from the Science Group Thermodata Europe (SGTE) compiled by Dinsdale [28]. R is the gas constant and T is the absolute temperature in Kelvin. ${}^{ex}G_m^{\varphi}$ is the excess Gibbs energy and ${}^{mag}G_m^{\varphi}$ is the magnetic contribution to the Gibbs energy of the phase φ, respectively. A_j and B_j are parameters to be optimized in this work. τ is the normalized temperature as expressed, $\tau = T/T_C^{\varphi}$, and T_C^{φ} is the Curie temperature of the phase φ. According the equation suggested by Hillert and Jarl [29], $g(\tau)$ is expressed as:

$$g(\tau) = \begin{cases} 1 - \frac{1}{M}\left[\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p} - 1\right)\left(\frac{\tau^3}{6} + \frac{\tau^9}{135} + \frac{\tau^{15}}{600}\right)\right] & (\tau > 1) \\ -\frac{1}{M}\left[\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500}\right] & (\tau < 1) \end{cases} \tag{5}$$

$$M = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p} - 1\right) \tag{6}$$

in which p is the determined by the structure of the phase φ (0.4 for bcc phase and 0.28 for other phases).

3.2. Intermetallic compounds

In the Fe-Tm and Fe-Ho binary systems, the intermetallic compounds, $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm, Fe_2Tm, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$, Fe_3Ho and Fe_2Ho, are considered to be stoichiometric compounds because of no experimental data for their composition ranges. The heat capacities of Fe_2Tm and Fe_2Ho were experimentally determined by Germano [23], respectively, and thus their molar Gibbs energies ($G_m^{Fe_pRE_q}$) are expressed as:

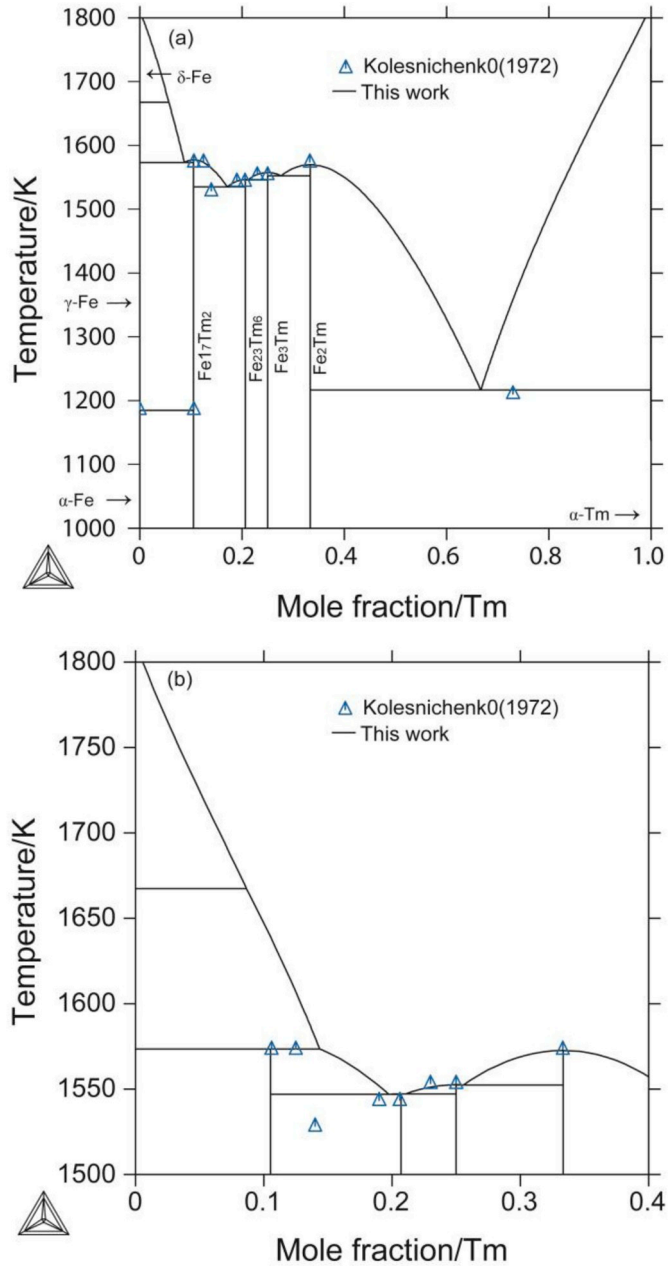

Fig. 2. Calculated phase diagram of the Fe-Tm system with the experimental data [17].

$$G_m^{Fe_pRE_q} = \begin{cases} C_{i_1} + D_{i_1}T + E_{i_1}T^3 + F_{i_1}T^4 + {}^{mag}G_m^{Fe_pRE_q} & (T < 120) \\ \left.\begin{array}{l} \frac{p}{p+q}{}^0G_{Fe}^{bcc} + \frac{q}{p+q}{}^0G_{RE}^{hcp} + C_{i_2} + D_{i_2}T + E_{i_2}T\ln T \\ +F_{i_2}T^2 + H_{i_2}T^{-1} + I_{i_2}T^3 + {}^{mag}G_m^{Fe_pRE_q} \end{array}\right\} & (T < 120) \end{cases} \quad (7)$$

where p and q were the value of stoichiometric number and the parameters C_i, D_i, E_i, F_i, H_i and I_i are to be optimized in this work.

On the other hand, due to lack of the experimental heat capacity, the molar Gibbs energies of $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$ and Fe_3Ho are expressed using the Neumann-Kopp rule as:

$$G_m^{Fe_pRE_q} = \frac{p}{p+q}{}^0G_{Fe}^{bcc} + \frac{q}{p+q}{}^0G_{RE}^{hcp} + J + KT + {}^{mag}G_m^{Fe_pRE_q} \quad (8)$$

where $G_m^{Fe_pRE_q}$ is the molar Gibbs energies of $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$ and Fe_3Ho and the parameters J and K are to be optimized.

Table 4
Invariant reactions in the Fe-Tm binary system.

Reaction	Type	T/K	Composition (x_{Tm}^L)	Reference
L + δ-Fe ↔ γ-Fe	Peritectic	1667.00	0.087	[17] (exp.)
		1667.00	0.067	[14] (cal.)
		1667.00	0.060	[16] (cal.)
		1667.50	0.086	This work (cal.)
L + γ-Fe ↔ $Fe_{17}Tm_2$	Peritectic	1573.00	0.125	[17] (exp.)
		1567.00	0.110	[14] (cal.)
		1580.00	0.125	[16] (cal.)
		1573.40	0.143	This work (cal.)
L↔ $Fe_{17}Tm_2$ + $Fe_{23}Tm_6$	Eutectic	1528.00	0.170	[17] (exp.)
		1533.00	0.176	[14] (cal.)
		1528.00	0.177	[16] (cal.)
		1540.00	0.193	This work (cal.)
L + Fe_3Tm ↔ $Fe_{23}Tm_6$	Peritectic	1543.00	0.207	[17] (exp.)
		1537.00	0.195	[14] (cal.)
		1537.00	0.205	[16] (cal.)
		1544.60	0.202	This work (cal.)
L + Fe_2Tm ↔ Fe_3Tm	Peritectic	1551.00	0.225	[17] (exp.)
		1549.00	0.242	[14] (cal.)
		1551.00	0.250	[16] (cal.)
		1552.50	0.255	This work (cal.)
L ↔ Fe_2Tm	Congruent	1573.00	0.333	[17] (exp.)
		1576.00	0.333	[14] (cal.)
		1573.00	0.333	[16] (cal.)
		1572.60	0.333	This work (cal.)
L ↔ Fe_2Tm + α-Tm	Eutectic	1210.00	0.730	[17] (exp.)
		1182.00	0.753	[14] (cal.)
		1214.00	0.672	[16] (cal.)
		1209.50	0.665	This work (cal.)
$Fe_{17}Tm_2$ + γ-Fe ↔ α-Fe	Peritectoid	1185.00	–	[17] (exp.)
		1185.00	–	[14] (cal.)
		1185.00	–	[16] (cal.)
		1185.00	–	This work (cal.)

It notes that ${}^{mag}G_m^{Fe_pRE_q}$ as shown in Equ. (7) and Equ. (8) is the magnetic contribution to the Gibbs energy and the formula is expressed by Equ. (4). The Curie temperature (T_C^{φ}) and Bohr magneton (β_0) of the intermetallic compounds in the Fe-Tm and Fe-Ho binary systems were used as given in Table 2 and Table 3, respectively.

4. Results and discussion

Based on the experimental information mentioned above in the literature, the thermodynamic parameters of all stable phases in the Fe-Tm and Fe-Ho binary systems were optimized using the PARROT module of Thermo-calc software developed by Sundman et al. [30]. Thermodynamic parameters in the Fe-Tm and Fe-Ho binary systems obtained finally were shown in Tables 2 and 3. The comparison of the calculated results with the experimental results was discussed in the following section.

4.1. The Fe-Tm binary system

Fig. 1 is the calculated Fe-Tm binary phase diagram with the calculated results [14,16]. As can be seen, four intermetallic compounds, $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm and Fe_2Tm, are stable at room temperature in this work, while some intermetallic compounds are decomposed abnormally at low temperature in the calculated results [14]. The comparison of the calculated Fe-Tm binary phase diagram with the experimental data reported by Kolesnichenko et al. [17] is shown in

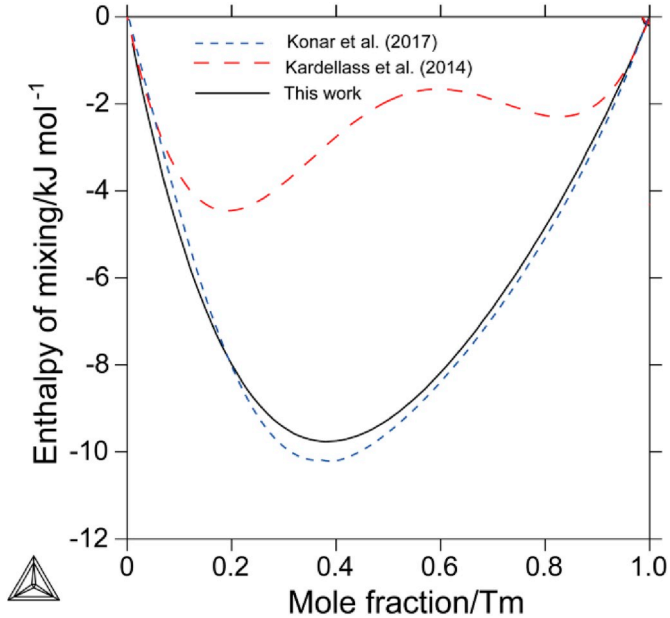

Fig. 3. Calculated enthalpy of mixing of liquid Fe-Tm alloys at 1833 K with the calculated results [14,16] (Reference states: liquid Fe and liquid Tm).

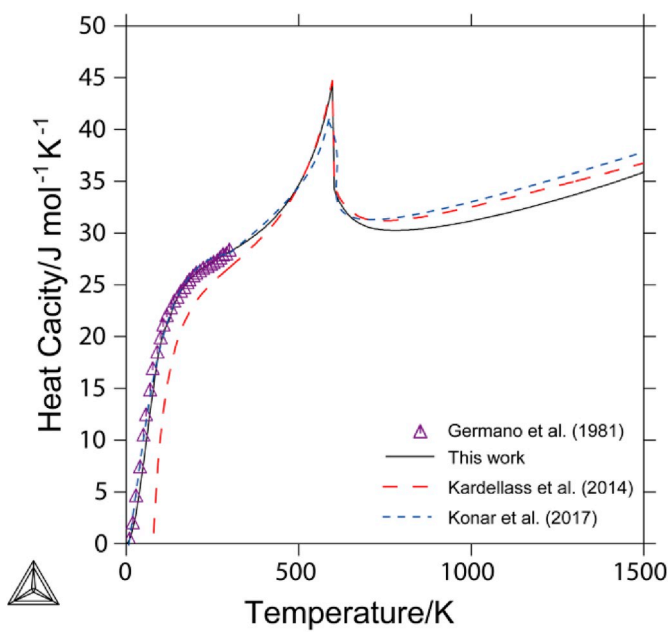

Fig. 4. Calculated heat capacity of Fe_2Tm with the experimental data [23] and the calculated results [14,16].

Fig. 2. The calculated liquidus and solidus are in good agreement with the experimental data [17]. The calculated temperatures and compositions of invariant reactions with the experimental data were summarized in Table 4. In this work, the calculated temperatures and compositions of two eutectic reactions, L ↔ $Fe_{17}Tm_2$ + $Fe_{23}Tm_6$ and L ↔ Fe_2Tm + α-Tm, are 1540.0 K and 19.3 at. % Tm at 1209.5 K and 66.5 at.% Tm, respectively. The calculated temperature of congruent melting of Fe_2Tm (L ↔ Fe_2Tm) is to be 1572.6 K and the calculated temperatures of four peritectic reactions, L + δ-Fe ↔ γ-Fe, L + γ-Fe ↔ $Fe_{17}Tm_2$, L + Fe_3Tm ↔ $Fe_{23}Tm_6$, and L + Fe_2Tm ↔ Fe_3Tm, are 1667.5 K, 1573.4 K, 1544.6 K, and 1552.5 K, respectively, which are consistent with the experimental data [17].

Fig. 3 is the calculated enthalpies of mixing of liquid Fe-Tm alloys at 1833 K with the calculated results [14,16]. Due to the lack of the experimental enthalpy of mixing of liquid Fe-Tm alloys, Konar et al. [16] estimated the enthalpy of mixing of liquid Fe-Tm alloys at 1833 K by means of the formula as$\Delta H_{mix,Fe-Tm} = \Delta H_{mix,Fe-Dy} + 0.75(\Delta H_{mix,Fe-Lu} - \Delta H_{mix,Fe-Dy})$. $\Delta H_{mix,Fe-Lu}$ and $\Delta H_{mix,Fe-Dy}$ are taken directly from the experimental results determined by Ivanov et al. [19] using the isoperibolic calorimetry under high purity He gas. The estimated results by Konar et al. [16] satisfies the tendency that the enthalpies of mixing of liquid Fe-RE alloys becomes more and more negative as the number of rare earth atoms increases. In this work, the calculated enthalpies of mixing of liquid Fe-Tm alloys are in good agreement with the calculated results [16], while there is a significant deviation between the calculated results [14] and the calculated results [16].

Fig. 4 is the calculated heat capacity of Fe_2Tm with the experimental data [23] and the calculated results [14,16]. The calculated heat capacity is consistent well with the experimental data [23], while the calculated results [14] is significantly different with the experimental data [23] in the low temperature range.

Fig. 5 is the calculated enthalpies of formation of intermetallic compounds in the Fe-Tm binary system at 298 K with the experimental data [24,25] and the calculated results [14,16,26,27]. In this work, the calculated enthalpies of formation of four intermetallic compounds, $Fe_{17}Tm_2$, $Fe_{23}Tm_6$, Fe_3Tm, Fe_2Tm, are −3.54 kJ/mol-atom, −8.02 kJ/mol-atom, −9.12 kJ/mol-atom and −11.63 kJ/mol-atom, respectively. The calculated enthalpy of formation of $Fe_{17}Tm_2$ is in good agreement with the experimental data determined by Gozzi et al. [24] and the calculated results [26,27], while the calculated enthalpy of formation of Fe_2Tm is slightly more negative than the experimental data measured by Meschel et al. [25]. It is noticed that the enthalpy of formation of Fe_2Tm measured by Meschel et al. [25] is more positive than that of $Fe_{17}Tm_2$ measured by Gozzi et al. [24]. Considering the congruent melting of Fe_2RE intermetallic compounds in the Fe-RE binary systems, the enthalpies of formation of Fe_2RE should be most negative. Therefore, the experimental data measured by Meschel et al. [25] is less reliable and the calculated result in this work is reasonable. It is very necessary that the enthalpies of formation of the intermetallic compounds in the Fe-Tm binary system should be measured in the further experiments.

4.2. The Fe-Ho binary system

Fig. 6 is the calculated Fe-Ho binary phase diagram in this work with the calculated results [15,16]. Fig. 7 is the calculated Fe-Ho binary phase diagram with the experimental data [18]. As can be seen, four intermetallic compounds, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$, Fe_3Ho and Fe_2Ho, are stable at room temperature in this work, while some intermetallic compounds are decomposed at low temperature in the calculated results [15]. The calculated Fe-Ho binary phase diagram in this work is in good agreement with the experimental data [18]. The calculated temperatures of four eutectic reactions (L ↔ $Fe_{17}Ho_2$ + $Fe_{23}Ho_6$ at 1590.3 K, L ↔ Fe_2Ho + α-Ho at 1146.7 K, L ↔ $Fe_{17}Ho_2$ + γ-Fe at 1612.3 K, L ↔ Fe_3Ho + Fe_2Ho at 1559.5 K) are in good agreement with the experimental data [18]. The temperatures of congruent melting of $Fe_{17}Ho_2$ (L ↔ $Fe_{17}Ho_2$) and $Fe_{23}Ho_6$ (L ↔ $Fe_{23}Ho_6$) were calculated to be 1615.6 K and 1605.2 K, while the calculated temperatures of three peritectic reactions, L + δ-Fe ↔ γ-Fe, L + $Fe_{23}Ho_6$ ↔ Fe_3Ho, L + β-Ho ↔ α-Ho are 1666.7 K, 1567.6 K and 1706.0 K, respectively, which are in good agreement with the experimental data [18]. Table 5 summarized the calculated temperatures and compositions of invariant reactions with the experimental data [18].

Fig. 8 is the calculated enthalpy of mixing of liquid Fe-Ho alloys at 1833 K with the calculated results [15,16]. The formula $((\Delta H_{mix,Fe-Ho} = \Delta H_{mix,Fe-Dy} + 0.25(\Delta H_{mix,Fe-Lu} - \Delta H_{mix,Fe-Dy})))$ was used by Konar et al. [16] to estimate the enthalpy of mixing of liquid Fe-Ho alloys according to the similar method mentioned above. As can be seen, the calculated results [15] show a significant deviation with

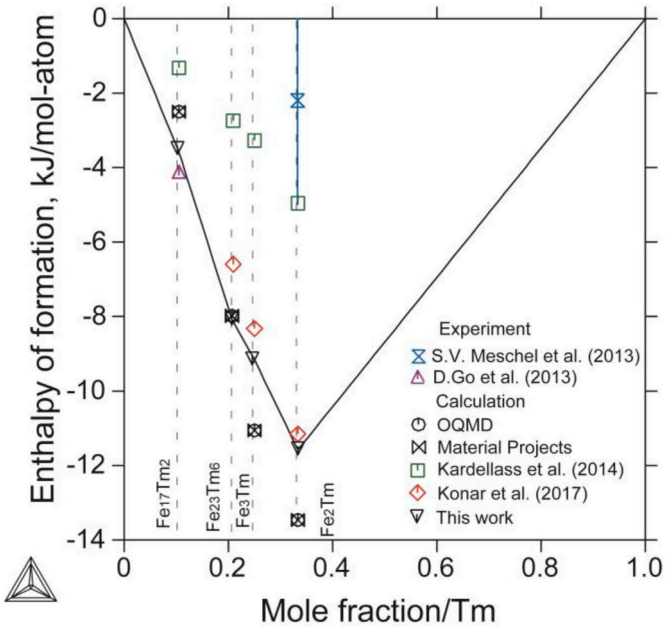

Fig. 5. Calculated enthalpies of formation of Fe-Tm intermetallic compounds at 298 K with the experimental data [24,25] and the calculated results [14,16,26,27]. (Reference states: α-Fe and α-Tm).

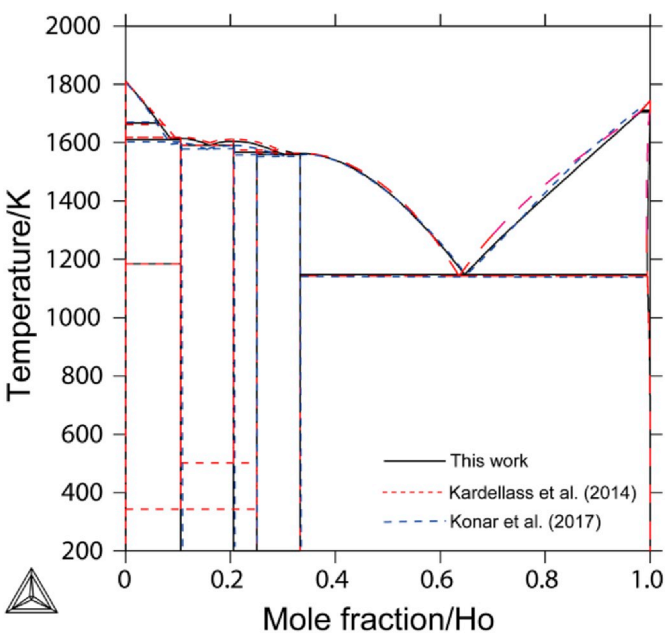

Fig. 6. Calculated phase diagram of the Fe-Ho binary system in this work with the calculated results [15,16].

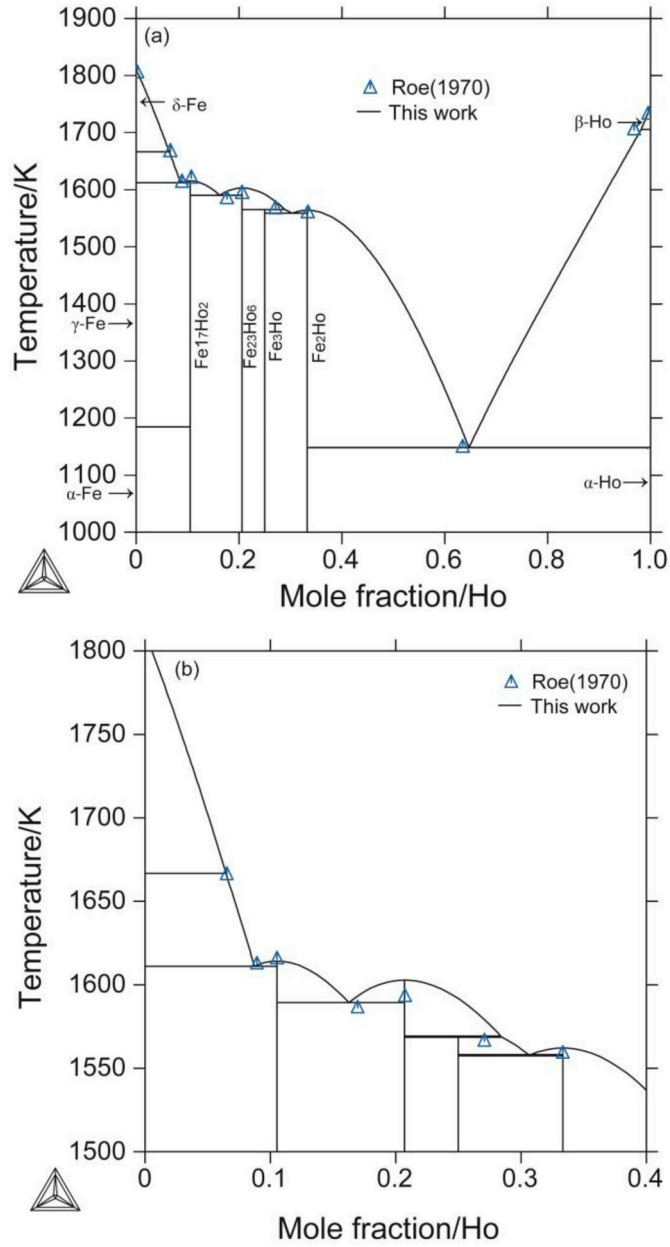

Fig. 7. Calculated phase diagram of the Fe-Ho binary system with the experimental data [18].

the calculated results [16], while the calculated results in this work are in good agreement with the calculated results [16].

Fig. 9 is the calculated heat capacity of Fe_2Ho with the experimental data [23] and the calculated results [15,16]. The calculated heat capacity of Fe_2Ho in this work is consistent well with the experimental data measured by Germano et al. [23].

Fig. 10 is the calculated enthalpies of formation of intermetallic compounds in the Fe-Ho binary system at 298 K with the experimental data [24,25] and the calculated results [15,16,26,27]. In this work, the calculated enthalpies of formation of four intermetallic compounds, $Fe_{17}Ho_2$, $Fe_{23}Ho_6$, Fe_3Ho and Fe_2Ho at 298 K are −2.36 kJ/mol-atom, −5.67 kJ/mol-atom, −7.32 kJ/mol-atom and −10.62 kJ/mol-atom, respectively. The calculated enthalpy of formation of $Fe_{17}Ho_2$ is in good agreement with the experimental data determined by Gozzi et al. [24] and the calculated results [26,27], while the calculated enthalpy of formation of Fe_2Ho is more negative than the experimental data measured by Meschel et al. [25]. It should be pointed out that the enthalpies of formation of Fe_2RE intermetallic compounds measured by Meschel et al. [25] are more positive than the reported experimental data and calculated results, which is not consistent with the tendency of the enthalpy of formation of Fe_2RE in the Fe-RE binary systems mentioned above. Hence, the calculated enthalpies of formation of intermetallic compounds in the Fe-Ho binary system are still accepted. The further experiments should be preformed to determine the enthalpies of formation of the intermetallic compounds in the Fe-Tm binary system.

In order to further discuss the validity of the calculated enthalpy of mixing of liquid Fe-Tm and Fe-Ho alloys in this work, the enthalpy of mixing of liquid Fe-RE (RE = Sm, Gd, Tb, Ho, Tm, Lu) alloys at 1833 K determined experimentally by Ivanov et al. [19] compares with the

Table 5
Invariant reactions in the Fe-Ho binary system.

Reaction	Type	T/K	Composition (x^L_{Ho})	Reference
L + δ-Fe ↔ γ-Fe	Peritectic	1667.00	0.067	[18] (exp.)
		1667.00	0.072	[15] (cal.)
		1666.70	0.066	This work (cal.)
L ↔ $Fe_{17}Ho_2$ + γ-Fe	Eutectic	1611.00	0.083	[18] (exp.)
		1608.00	0.096	[15] (cal.)
		1608.00	0.082	[16] (cal.)
		1612.30	0.086	This work (cal.)
L ↔ $Fe_{17}Ho_2$	Congruent	1616.00	0.105	[18] (exp.)
		1613.00	0.105	[15] (cal.)
		1614.00	0.105	[16] (cal.)
		1615.60	0.105	This work (cal.)
L ↔ $Fe_{17}Ho_2$ + $Fe_{23}Ho_6$	Eutectic	1585.00	0.178	[18] (exp.)
		1591.00	0.163	[15] (cal.)
		1588.00	0.160	[16] (cal.)
		1590.30	0.161	This work (cal.)
L ↔ $Fe_{23}Ho_6$	Congruent	1604.00	0.206	[18] (exp.)
		1603.00	0.206	[15] (cal.)
		1605.00	0.206	[16] (cal.)
		1605.20	0.206	This work (cal.)
L + $Fe_{23}Ho_6$ ↔ Fe_3Ho	Peritectic	1566.00	0.271	[18] (exp.)
		1569.00	0.289	[15] (cal.)
		1573.00	0.281	[16] (cal.)
		1567.60	0.287	This work (cal.)
L ↔ Fe_2Ho + Fe_3Ho	Eutectic	1561.00	0.349	[18] (exp.)
		1554.00	0.317	[15] (cal.)
		1557.00	0.314	[16] (cal.)
		1559.50	0.303	This work (cal.)
L ↔ Fe_2Ho	Congruent	1563.00	0.333	[18] (exp.)
		1555.00	0.333	[15] (cal.)
		1559.00	0.333	[16] (cal.)
		1565.00	0.333	This work (cal.)
L ↔ Fe_2Ho +α-Ho	Eutectic	1148.00	0.634	[18] (exp.)
		1146.00	0.632	[15] (cal.)
		1148.00	0.665	[16] (cal.)
		1146.70	0.637	This work (cal.)
L + β-Ho ↔ α-Ho	Peritectic	1709.00	–	[18] (exp.)
		1709.00	0.983	[15] (cal.)
		1706.00	0.984	This work (cal.)
$Fe_{17}Ho_2$ + γ-Fe ↔ α-Fe	Peritectoid	1185.00	–	[18] (exp.)
		1185.00	–	[15] (cal.)
		1185.00	–	[16] (cal.)
		1184.80	–	This work (cal.)

assessed results [12,13] in the reported literature as given in Fig. 11. With the increase of atomic number of rare earth metals from La to Lu, the enthalpies of mixing of liquid Fe-RE alloys become much more negative and show the regular change from −4 kJ/mol-atom to −12 kJ/mol-atom. The systematic tendency of the enthalpy of mixing of liquid Fe-RE alloys should be considered to optimize thermodynamic parameters of liquid phases in the Fe-RE binary systems. As can be seen in Fig. 11, in this work, the calculated enthalpies of mixing of liquid Fe-Tm and Fe-Ho alloys agree well with the systematic change and thus are reasonable. On the other hand, Fig. 12 presents the calculated enthalpies of formation of intermetallic compounds in the Fe-RE (RE = Sm, Gd, Tb, Ho, Tm, Lu) binary systems at 298 K with the calculated results [12,13]. In general, the enthalpies of formation of Fe_2RE compounds in the Fe-RE binary systems are most negative. It is worth noting that the experimental data determined by Meschel et al. [25]

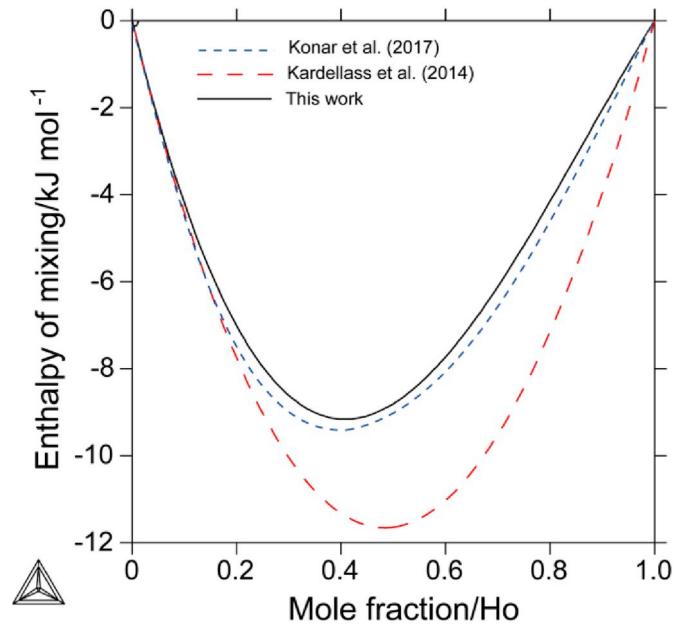

Fig. 8. Calculated enthalpy of mixing of liquid Fe-Ho alloys at 1833 K with the calculated results [15,16] (Reference states: liquid Fe and liquid Ho).

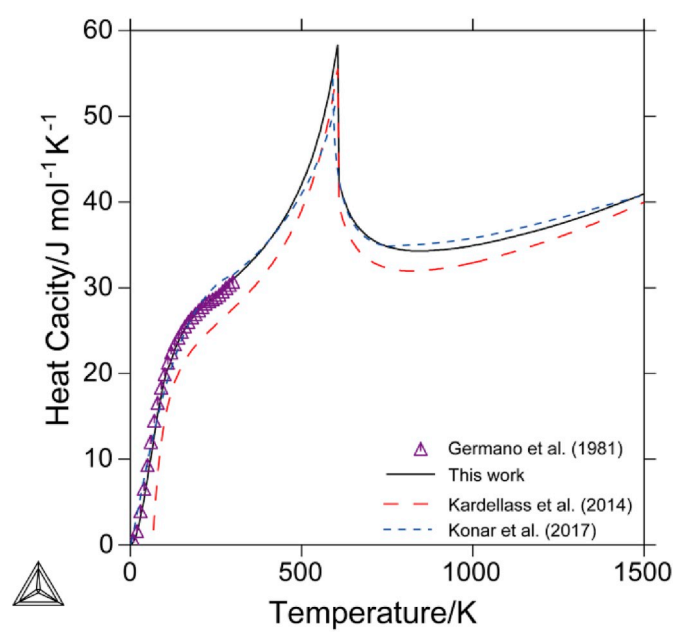

Fig. 9. Calculated heat capacity of Fe_2Ho with the experimental data [23] and the calculated results [15,16].

was more positive than the calculated results in this work. Further experiments would be still performed to measure thermodynamic properties (e.g. enthalpy of mixing and enthalpy of formation) of the Fe-RE binary system.

5. Conclusions

On the basis of the critical review of the previous optimizations and experimental information on phase diagram data and thermodynamic properties in the published literature, thermodynamic re-assessment of the Fe-Tm and Fe-Ho binary systems was performed using the CALPHAD method. Self-consistent and reliable thermodynamic parameters of the Fe-Tm and Fe-Ho binary systems were obtained finally, which can reproduce satisfactorily most of the experimental data

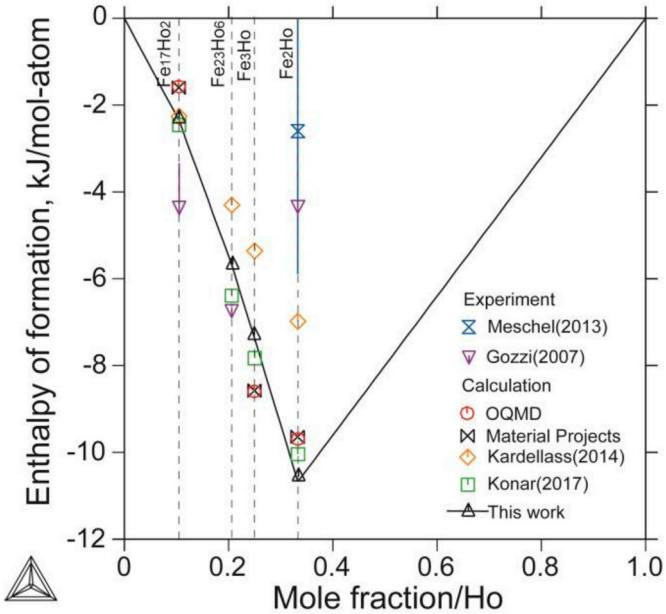

Fig. 10. Calculated enthalpies of formation of Fe-Ho intermetallic compounds at 298 K with the experimental data [24,25] and the calculated results [15,16,26,27]. (Reference states: α-Fe and α-Ho).

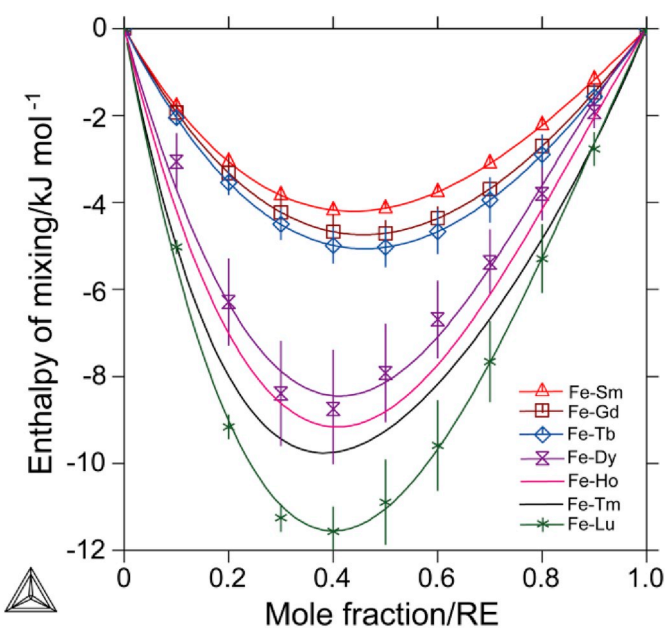

Fig. 11. Calculated enthalpy of mixing of liquid Fe-RE alloys at 1833 K with the experimental data [19] and the calculated results [12,13] (Reference states: liquid Fe and liquid RE).

including phase equilibria and thermodynamic properties. It is fundamental to develop the compatible and available thermodynamic database of the RE-Fe-B-based alloys, which is very useful to investigate the relations between alloy composition, microstructure and magnetic properties of novel RE-Fe-B-based permanent magnets.

Conflicts of interest

The authors of the manuscript 'Thermodynamic re-assessment of the Fe-Tm and Fe-Ho binary systems' declare no conflict of interest.

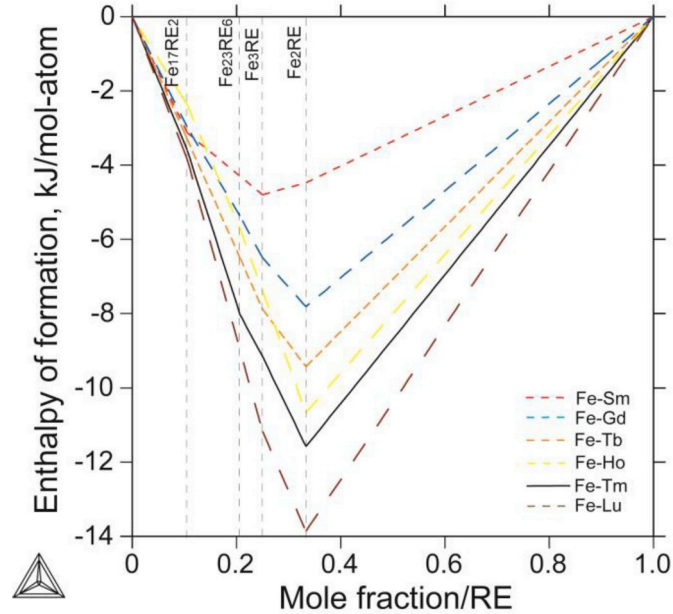

Fig. 12. Calculated enthalpies of formation of Fe-RE (RE = Sm, Gd, Tb, Ho, Tm, Lu) intermetallic compounds at 298 K with the calculated results [12,13]. (Reference states: α-Fe and α-RE).

Acknowledgements

This work was supported financially by the National Natural Science Foundation of China (51761008, 51461013), the National Key Research and Development Program of China (2016YFB0700901), the National Basic Foundation of China (2014CB643703), the Guangxi Natural Science Foundation (2016GXNSFDA380015, 2016GXNSFGA380001), the Guangxi Project of Science and Technology (201723031 CE) and Guangxi Key Laboratory of Information Materials, Guilin University of Electronic Technology, China (171005-Z).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2019.101646.

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